

Web vs. Internet

- **Web (World Wide Web):** "It is a network of web pages that are found using a link address. It is called the "web" based on its interconnected nature, designed to easily get various resources."
- The web allows access to an array of documents that are interconnected through hypertext or hypermedia links, i.e., hyperlinks and electronic connections that link similar pieces of information, allowing the user to smoothly transition from one page to another.

Brief History of the Web

- **Tim Berners-Lee:** "He is a British scientist who invented the Web, also called World Wide Web (WWW), in 1989 while working at CERN."
- The fundamental idea was to combine the growing technologies of computers, data networks, and hypertext into a powerful and simple global information system use.
- By the end of 1990, he had the first Web server and browser running at CERN while developing the code for his Web server on a NeXT computer, putting up a hand-written label in red ink that says "This machine is a server. DO NOT POWER IT DOWN!!"
- **The First-Ever Website:** The first web page went live on August 6, 1991, addressed as [\[http://info.cern.ch/hypertext/WWW/TheProject.html\]](http://info.cern.ch/hypertext/WWW/TheProject.html)(<http://info.cern.ch/hypertext/WWW/TheProject.html>).
- **Extension and Going Global:** In March 1991, the WWW software became available to colleagues that use CERN computers, and by 1993, the National Center for Supercomputing Applications (NCSA) released versions for PC and Macintosh environments.

The Internet

- "It is a massive interconnection of networks connecting millions of computers globally, thus forming a network wherein any computer can communicate with other computers connected to it."
- It is also a network of physical cables including copper telephone wires, TV cables, and fiber optic cables, while wireless connections such as Wi-Fi and 3G/4G rely on these physical cables to access the Internet.
- The process of internet connection starts with entering a website where the computer sends a request over the wires to a server where websites are stored, and once the request arrives, the server recovers the website and sends the retrieved data back to the computer.

Brief History of the Internet

- **The 1960s:** Joseph Carl Robnett Licklider started an idea for a global computer network in **1962**, and Roberts published a plan for Advanced Research Projects

Agency Network (ARPANET), making it a standard for all military computer networking.

- **The 1980s:** The PhoneNet system was founded in **1982**, and Paul Mockapetris, Jon Postel, and Craig Partridge developed the Domain Name System, with [symbolics.com](https://www.symbolics.com) registered as the first domain in **1984**.
- **The 1990s:** ARPANET was decommissioned, and Tim Berners-Lee and his colleagues developed hypertext markup language (HTML) and the uniform resource locator (URL). In **1995**, Microsoft launched Windows 95, online businesses like Amazon, Yahoo, and eBay followed suit, and Java was created.
- **The 2000s:** Saw the rise and burst of the dot-com bubble (Internet bubble) between **1995-2000**, Google's meteoric rise to domination in the search engine market, and the emergence of Wi-Fi and mobile internet devices like smartphones.

Web vs. Internet (Summary Distinction)

- "Simply put, the **web is an application built on the internet**. The **internet is the infrastructure that connects computer networks**, while the **web is a system of information accessible through the Internet**."
- Not everything available on the internet requires the web; for example, sending an email does not rely on the web or a web browser as it uses SMTP (Simple Mail Transfer Protocol).

How the Web and the Internet Changed the World

- **Communication:** Messages sent via mail that once took days or weeks now use email and social media sites for instant communication, though this also creates an environment of careless communication and impulsive messaging.
- **Information:** The web and the internet gave the world access to various information resources and search engine sites like Google, though a downside is the spread of false information or "fake news" with barely any fact-checking.
- **Entertainment:** The industry has been impacted through social media, gaming websites (Twitch and YouTube Gaming), digitalized newspapers and magazines, and streaming services (Netflix, Disney+, Spotify, and Apple Music).
- **Education:** Played a significant role during lockdowns caused by the COVID-19 pandemic by forcing educational systems to adapt to an online learning model, utilizing video conferencing tools and online learning platforms like the NEO eLMS portal.